

AI Express Hackathon: Student Instruction & Submission Sheet

Assessment Weightage: 10 Marks

Duration: 90 Minutes

Team Size: 3 Students per group

Scope: Units 1 to 4 (AI Foundations, Problem-Solving Agents, Logical Agents, First-Order Logic)

1. Hackathon Overview & Rules

Every team must build an autonomous agent that navigates a designated grid environment based on one of the assigned syllabus tracks.

- **Mandatory Requirement:** Your agent **must visually navigate its environment (via Tkinter, Pygame, Matplotlib, or live ASCII terminal rendering)** and log its decision-making process in real time.
- **GitHub Requirement:** All source code, setup scripts, and visual assets must be pushed to a public or shared **GitHub Repository** before the 90-minute deadline.
- **Collaboration:** All 3 team members must contribute actively with visible commit histories on GitHub.
- **Strict Time Limit:** Code execution, GitHub push, screen recording, and PDF summary must all be completed within the 90-minute window.

2. Submission Deliverables

You are required to submit **three components** before the deadline:

1. **GitHub Repository Link:** Public repository link containing your agent code, README.md, and execution instructions.
2. **Screen Recording Video:** Uploaded directly (.mp4 file) OR linked inside your GitHub README (YouTube / Loom / Drive link).
3. **Technical Summary Sheet:** 1-Page .pdf uploaded to the portal and saved in your GitHub root directory as SUMMARY.pdf.

3. Video Submission Guidelines (60–90 Seconds)

Your video serves as the primary proof of your agent's live autonomous execution.

- **Length:** Strictly **60 to 90 seconds** long.
- **Screen Setup:** Show a **split-screen or dual window**: one window displaying the visual grid simulation where the agent moves, and the other displaying the live console log.
- **Required Content in Video:**

- **First 15 Seconds:** Brief voiceover introducing team members, GitHub repo link, selected Track, and environment setup.
- **Middle 45–60 Seconds:** Live demonstration of the moving agent reacting to environmental states (recalculating paths in A*, resolving Knowledge Base updates in Propositional/FOL logic, or dynamic replanning).
- **Final 15 Seconds:** Brief highlight of performance metrics (time taken, path cost, or nodes expanded).

4. Technical Summary Sheet Format (1-Page PDF)

Keep your documentation concise using the following 4 sections:

1. **Header & Team Info:** Course Code, Group ID, Member Names, Selected Track, and GitHub Repository URL.
2. **PEAS Framework Matrix:** Performance Measure, Environment, Actuators, and Sensors.
3. **Core Algorithmic Formulation:** State space definition, initial state, goal test, path cost, and explicit heuristic equations $h(n)$ or logic rules used.
4. **Complexity Analysis:** Theoretical Time and Space complexity (O notation) compared with observed execution metrics.

5. Evaluation & Marking Breakdown (10 Marks)

Evaluation Parameter	Marks	Criteria
Live Agent Movement (Video)	4 Marks	The video clearly demonstrates real-time visual movement and autonomous decision-making.
Algorithmic Correctness	3 Marks	Proper implementation of search heuristics (A*), Propositional resolution, FOL engines, or Dynamic Replanning.
GitHub Code Quality & Team Commits	2 Marks	Clean codebase pushed to GitHub, structured repository, and active commit history from all 3 team members.
Technical Summary Sheet	1 Mark	Complete a 1-page PDF detailing GitHub URL, PEAS framework, logic formulas, and complexity analysis.